Sime Darby Property University Hackathon 2026 · Universiti Malaya

Team HEXA

Five Universiti Malaya students in Data Science, Actuarial Science, and AI—building practical car-park operations for Sime Darby Property.

B1-01

SE

HEXA 01

Leow Shen En

Team lead, product & UI/UX

Data Science

B1-02

PC

HEXA 02

Pohyi Chong

Forecasting & ML models

Data Science

B1-03

IT

HEXA 03

Isaac Teo

Data analysis & simulation

Data Science

B1-04

JS

HEXA 04

Lee Jia Shern

Business case & finance

Actuarial Science

B1-05

ZL

HEXA 05

Choong Zhuo Lin

AI engineering & prototype

Artificial Intelligence

Value Proposition

01

Member benefit

Eligible members reserve access to pooled Priority Zone capacity right beside the B1 lifts.

02

Automated control

Entry and exit plate cameras protect active bookings and release capacity the moment a car leaves.

03

Auditable proof

SDP tracks utilisation, fulfilment reliability, shopper satisfaction, and repeat-visit uplift in one view.

Members	A practical weekend perk they feel on every trip to the mall.
Mall operations	One live, auditable log of arrivals, departures, no-shows, and exceptions.
Loyalty team	A differentiated perk that rewards high-frequency spenders without discounting retail goods.
SDP portfolio	A blueprint ready to replicate across SDP malls once KL East clears its value gate.

Direct income: eligible members pay RM4 on weekdays or RM5 at peaks. Sponsorship, passes and external licensing are separate options.

Existing CCTV can read free bays accurately enough for a pilot.

99.05%

of spots read correctly, on 4 cameras the AI had never seen before (78,270 photos from a public car park dataset).

99.1%
accurate car count per camera. Our trial target is 95%.

96.1%
of camera views counted within one car

1 ms
to check each spot, even on a normal laptop

- Sunny 99.2%
- Overcast 99.3%
- Rainy 98.6%

To be honest: this was an outdoor car park. Basements are darker, so we will test again on B1 at KL East Mall during the trial.

We tested every option across 1,000 simulated busy Saturdays.

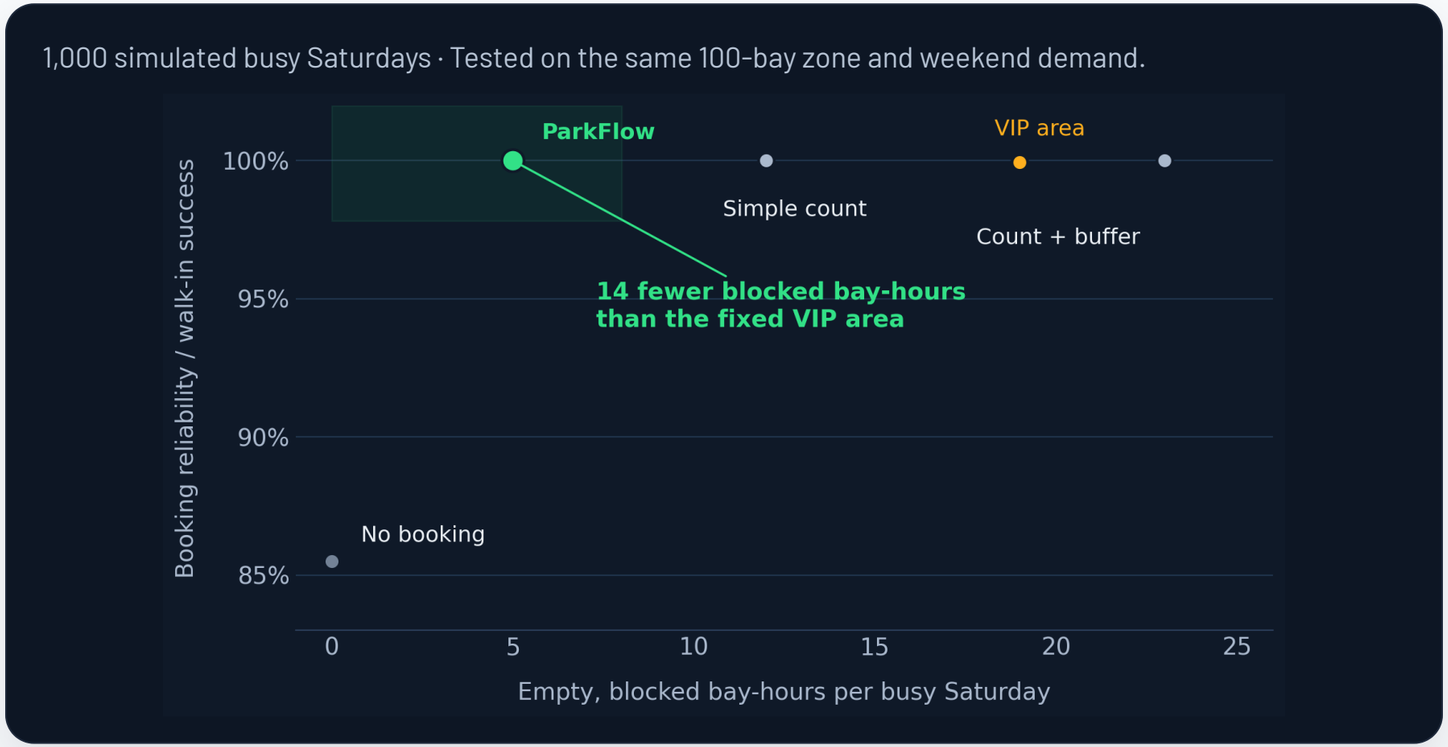
We can't wait for 1,000 real Saturdays, so we built them in a computer and tried every option on exactly the same days.

1. **Create a busy day**
   Crowds at lunch and after 5 pm, most shoppers staying about 1.5 hours, 60 bookings, and some people who never show up.
2. **Try every option on it**
   No booking at all, a fixed VIP area, a simple counting rule, and ParkFlow, on the same 100 spots.
3. **Repeat 1,000 times**
   Then check: did everyone who booked get a spot? How long did spots sit empty while other drivers were turned away?



1,000 busy Saturdays, one dot each

100% of booked members got a bay—with 74% less idle space.



100%

of booked members got a bay with ParkFlow, compared with 86% without booking control.

5 vs 19

empty, held bay-hours per Saturday—freeing 14 extra bay-hours compared with a fixed VIP zone.

95% vs 91%

peak zone utilisation between 5:00 pm and 9:00 pm.

Even when everything goes wrong at once, 99.6% of bookings hold.

Baseline | Stays 20% longer | Higher show-up rate | 10% ignore sign | All three at once

Control method	Bookings honoured	Idle held bay-hours
Fixed VIP zone	99.94%	143 h
Counting rule + 5-bay buffer	100%	127 h
Simple counting rule	100%	62 h
ParkFlow	100%	33 h

99.6%

of booked members still get a bay when longer stays, higher turnouts, and sign violations happen at the same time.

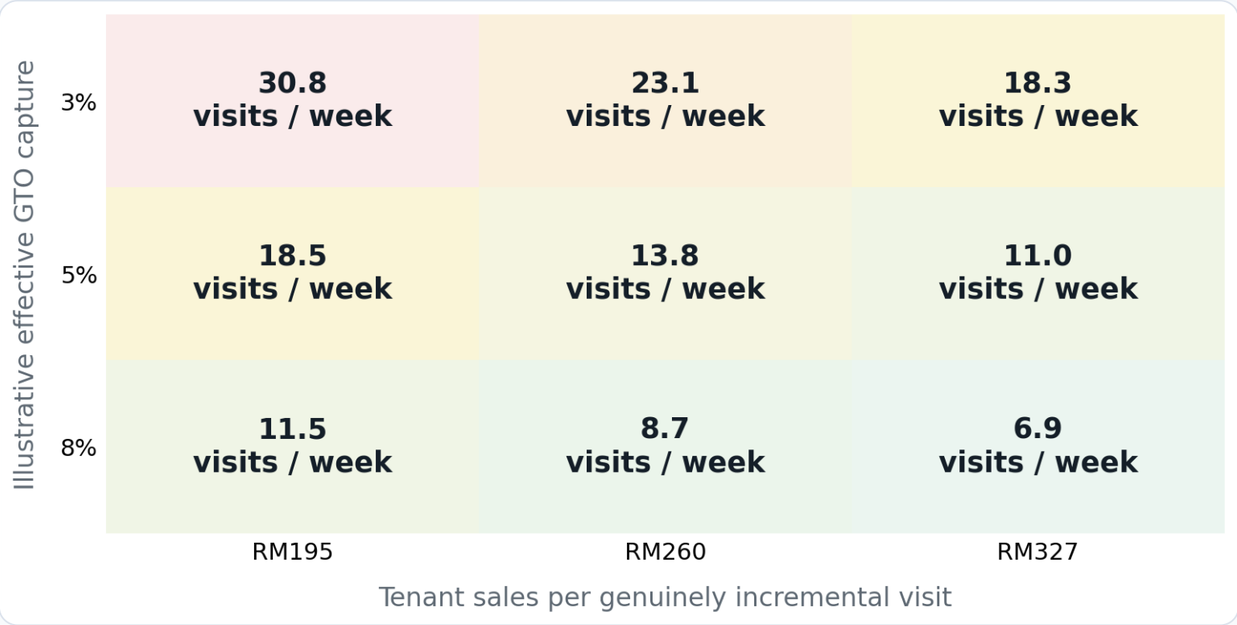
93.4%

booking fulfilment for a fixed VIP zone under the exact same stress test.

A strict counting rule also protects bookings, but locks up nearly four times as many empty bay-hours.

Retention Sensitivity

An illustrative RM9,360 annual contribution requires 7–31 additional visits a week, depending on spend and mall capture.



Incremental weekly visits needed across basket sizes and effective GTO capture rates, calibrated with SDP finance during co-design.

Observed Basket size	Malaysian mall benchmarks place average trip spend between RM195 and RM327.
Assumed Mall capture	Modeled across a 3% to 8% effective GTO rent and ancillary capture range.
Measured Pilot uplift	Track net additional member visits rather than raw app clicks or reservations.
Decision Rollout gate	Scale across the portfolio once measured retention clears SDP's return hurdle.

Payback: validate SDP's rental terms and measure additional visits before estimating capital recovery.

APPENDIX A1 · PILOT SCORECARD

Six clear questions decide whether ParkFlow scales after Week 8.

Dimension	Pre-pilot baseline	Pilot measurement	Pass condition
Member promise	Current weekend parking search experience	Share of booked arrivals that enter and find an open bay	Fulfilment reliability meets or exceeds the agreed SLA
Member demand	Peak member footfall and B1 occupancy	Active users, booking conversion, and repeat reservations	Consistent uptake from target loyalty tiers during peak hours
Loyalty value	Baseline shopper satisfaction and visit frequency	Pre/post member survey and tracked revisit frequency	Measurable lift in repeat visits among participating members
Operations	Current marshal workload and ramp incidents	Manual overrides, gate disputes, and staff minutes per day	Runs smoothly within existing concierge and marshal staffing
Data & privacy	Existing gate and CCTV counter accuracy	Reconciled entry, exit, occupancy, and booking logs	Audit-ready records compliant with the 2024 PDPA amendments
Unit economics	SDP loyalty retention budget and hurdle rate	Verified retention uplift vs. quoted build and run costs	Projected return clears SDP's portfolio investment threshold

Pass thresholds are locked in with SDP management prior to Week 1 so pilot evaluation stays objective.

APPENDIX A2 · PILOT COST BREAKDOWN

Every pilot cost line is transparent and tied to a site-survey check.

Cost item	Planning estimate	Basis of estimate	Pre-approval action
Zone LPR cameras (installed)	RM12,000	Commercial list-price estimate	Confirm B1 entry/exit sightlines and installer quote
Dynamic LED zone sign	RM16,300	Indicative supplier quotation	Confirm dimensions, brightness, mounting, and power feed
Two local edge AI boxes	RM2,340	Retail hardware pricing (Jetson Orin Nano)	Confirm comms room enclosure and thermal specs
Blind-spot aisle cameras	RM5,000	Provisional allowance pending site walk	Verify exact camera count during KL East B1 survey
Cabling & electrical works	RM7,500	Engineering installation estimate	Obtain contractor quote for PoE runs and conduit
Production software integration	RM60,000	Prototype-to-production engineering scope	Confirm KL East Mall app API, SSO, and DPO sign-off
Year 1 operating & cloud cost	RM10,500	Cloud hosting, messaging, and hardware upkeep	Validate message volume and support SLA with SDP IT
Total pilot planning envelope	**RM113,640**	Base estimate before contingency	Finalise contingency buffer once vendor quotes are in

APPENDIX A3 · EVIDENCE LEDGER

What is already proven vs. what the KL East pilot will validate in the field.

Area	Current status	Evidence in hand	Validated during the KL East pilot
Higher spend by frequent shoppers	Observed	Vase.ai Malaysia Shopping Mall Report 2026 (n = 500)	Actual spend and revisit uplift among KL East members
Booking reliability under stress	Simulated	1,000 simulated busy Saturdays across 5 stress scenarios	Live B1 gate fulfilment and driver compliance rates
60-minute occupancy forecasting	Back-tested	48,112 forecasts tested across 30 real-world car parks	Calibration to KL East cinema and dining turnover peaks
RM113.6k pilot budget	Costed	Component list prices and engineering build scope	Binding local contractor and system-integrator quotes
Member retention economics	Pilot test	Weekly visit sensitivity matrix across GTO capture rates	Matched pre/post member visit frequency with SDP Loyalty
Portfolio & platform rollout	Stage-gated	Three-horizon expansion roadmap	Repeatability across Melawati, Elmina, and KLGCC malls

Published market data · Simulated, back-tested, or costed · Validated live in the KL East pilot

Priority Zone Access Rules

Member eligibility	Both modes remain members-only. A walk-in is an eligible member without a booking.
Admission control	Before a walk-in enters, verified empty capacity must cover outstanding arriving bookings, the safety buffer and that new admission.
Mode switching	Forecast demand guides the switch to Priority mode or a return to Normal mode. A predicted departure cannot create bookable capacity.
Existing parked members	Priority mode restricts new walk-in entry. Members already inside stay until they leave.
Pilot validation	Agree buffer size, switching thresholds and operating responsibilities before launch. Local forecasts and the switching policy require field validation.